

FUNCTIONAL SPECIFICATION DOCUMENT

ACP Digital Lending Platform — Corporate Banking Implementation

Client: Fidelity Bank Nigeria PLC

Document Reference: FBN-FSD-2026-001

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Classification: Confidential

EXECUTIVE SUMMARY

Fidelity Bank Nigeria PLC ("the Bank") is the 6th largest bank in Nigeria by total assets, with NGN 4.2 trillion in total assets as of December 2025, over 8.5 million customers, and 250 branches across Nigeria's 36 states. The Bank's Corporate Banking division manages a loan portfolio of NGN 1.1 trillion, comprising approximately 3,200 active corporate credit facilities.

The Bank's current credit origination process is entirely paper-based, relying on manual document collation, email-based credit memo routing, and spreadsheet-based credit scoring. The average turnaround time from application submission to credit committee decision is 21 business days, significantly above the industry benchmark of 7-10 days for peer banks using digital platforms.

The Board of Directors approved the Strategic Digital Transformation Program (SDTP) in February 2026, with the ACP implementation as the flagship initiative. This Functional Specification Document defines in detail the business requirements, functional specifications, data flows, user stories, and acceptance criteria for the five modules to be implemented on the Axe Credit Portal (ACP) platform.

1. PROJECT CONTEXT AND BACKGROUND

1.1 Current State Assessment

The Bank conducted a comprehensive process audit in January 2026 (Ernst & Young, Report ref: EY-FBN-2026-01) that identified the following critical pain points:

Credit Origination:

- Average 21 days application to approval
- 34% of applications require rework due to incomplete documentation
- No real-time visibility on application status for Relationship Managers
- Credit memos created manually in Microsoft Word, prone to formatting errors and version

control issues

- Scoring performed in Excel spreadsheets with no audit trail of changes

Portfolio Monitoring:

- Manual monthly portfolio review by Credit Risk team
- Early warning signals identified reactively (after covenant breach)
- No automated monitoring of external triggers (industry news, credit bureau)
- Portfolio health dashboard updated monthly, not real-time

Regulatory Compliance:

- CBN examination in November 2025 cited deficiencies in credit file documentation standards
- Audit trails for credit decisions incomplete for 23% of reviewed files
- Covenant monitoring manually tracked in Excel, last updated irregularly

1.2 Strategic Objectives

The ACP implementation must deliver the following measurable outcomes within 6 months of go-live:

KPI	Current	Target	Measurement
Application to decision turnaround	21 days	≤7 days	Average calendar days
Documentation completeness at submission	66%	≥95%	% complete applications
Scoring audit trail compliance	77%	100%	CBN exam metric
Early warning detection lead time	Reactive	≥60 days	Days before default
Credit officer productivity	4 files/month	12 files/month	Files processed per RM

1.3 Regulatory Framework

All configurations must comply with the following CBN regulations:

- CBN Prudential Guidelines for Banks and Discount Houses (2020)
- CBN Guidelines on Know Your Customer (KYC) (2023 Revision)
- CBN Risk-Based Supervision Framework
- Nigeria Data Protection Regulation (NDPR) 2019
- Financial Reporting Council of Nigeria (FRCN) Standards

- Basel III implementation guidelines for Nigerian banks (CBN Circular BSD/DIR/GEN/LAB/12/058)
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2. MODULE 1 — KYC AND CORPORATE ONBOARDING

2.1 Business Context

The Bank currently collects KYC documentation through a combination of physical forms, email submissions, and manual data entry into the core banking system (Finacle). The process involves 7 handoffs between departments and takes an average of 8 days from initial client contact to account readiness.

The CBN 2023 KYC revision introduced enhanced due diligence requirements for corporate customers, including mandatory beneficial ownership registers and the identification of Politically Exposed Persons (PEPs) within corporate structures.

2.2 Detailed Functional Requirements

2.2.1 Corporate Onboarding Screens

FR-KYC-001: Multi-Step Onboarding Form The system shall provide a structured, multi-step digital onboarding form for corporate clients organized into the following sections:

Step 1: Company Identification

- Legal name (as registered with CAC)
- Registration type (Limited Liability Company, Public Company, Partnership, Sole Proprietorship)
- CAC Registration Number (format validation: RC followed by 6-8 digits)
- Date of incorporation
- Registered office address
- Operating address (if different)
- Tax Identification Number (TIN) — format: 8 digits
- BVN of Directors (Bank Verification Number — 11 digits each)
- Nature of business (NAICS code selection from dropdown)
- Years in operation
- Number of employees (bands: 1-10, 11-50, 51-200, 201-500, 500+)

Step 2: Financial Profile

- Latest 3 years of audited financial statements (upload)
- Latest management accounts (not older than 6 months, upload)
- Projected financial statements (next 2 years, upload)

- Existing bank accounts and facilities (schedule)
- Annual turnover (latest year, NGN)
- Total existing debt obligations (NGN)
- Net worth (NGN)

Step 3: Ownership and Management

- Directors list (name, BVN, nationality, shareholding %)
- Beneficial owners (any individual owning $\geq 5\%$ directly or indirectly)
- PEP declarations for each director and beneficial owner
- Organizational chart upload
- Board resolution authorizing credit application (upload)

Step 4: Compliance and Documentation

- AML questionnaire (12 questions, mandatory responses)
- Source of funds declaration
- Tax clearance certificate upload
- CAC Form C02 or C07 (for companies)
- Utility bill (for address verification, not older than 3 months)
- Passport photographs of directors
- Any existing credit facilities with other banks (schedule with amounts and tenors)

FR-KYC-002: Field Validation Rules

- All mandatory fields must be enforced with inline validation messages
- CAC number format: "RC" followed by 6-8 numeric digits (e.g., RC123456)
- TIN format: 8 consecutive digits
- BVN format: 11 digits, must pass Luhn check
- Phone numbers: Nigerian format (+234 followed by 10 digits)
- Amounts: NGN currency, minimum NGN 0, maximum NGN 999,999,999,999
- Dates: DD/MM/YYYY format, incorporation date must be in the past
- Email addresses: standard RFC 5322 validation

FR-KYC-003: Document Management

- Accepted formats: PDF only
- Maximum file size: 25MB per document
- Multiple file upload supported (up to 10 files per section)
- Document metadata captured: upload date, uploaded by, document type

- Documents stored with version control (replacement creates new version)
- Expiry date tracking for time-sensitive documents (tax clearance, management accounts)

FR-KYC-004: Application Status Tracking

- Real-time status visible to Relationship Manager: Draft, Submitted, Under Review, Approved, Rejected
- Estimated completion percentage displayed during form fill
- Auto-save every 3 minutes during data entry
- Last edited timestamp displayed prominently

FR-KYC-005: AML/CFT Integration Hook

- On submission, system shall call the Bank's Fircosoft screening API
- Screening results to be captured and stored in the credit file
- Applications with positive AML hits to be flagged for manual review by Compliance
- Compliance override capability with mandatory justification and supervisor approval

FR-KYC-006: Data Pre-Population

- If client exists in core banking system (Finacle), pre-populate available data fields
- Pre-populated fields marked with indicator, editable
- Change log maintained for any modifications to pre-populated data

2.3 User Stories

ID	As a	I want to	So that
US-KYC-01	Relationship Manager	complete a client onboarding form digitally	I eliminate paper and reduce errors
US-KYC-02	Credit Operations	receive complete files automatically	I stop chasing missing documents
US-KYC-03	Compliance Officer	see AML screening results inline	I take action without accessing Fircosoft directly
US-KYC-04	Client	upload documents from my phone	I don't need to physically visit a branch
US-KYC-05	Credit Risk Manager	see KYC completeness score	I know which files need attention

2.4 Screen Wireframe Descriptions

The onboarding interface shall consist of:

- Progress indicator at the top showing all 4 steps with completion status
- Left sidebar summary showing sections and their completion status
- Main content area for the active step
- Navigation buttons: "Save Draft", "Previous", "Next", "Submit"
- Help tooltip on every complex field (?)
- Right panel: document checklist with upload status indicators

2.5 Business Rules Configuration

The following business rules must be configured in ACP's BRM (Business Rules Management) engine:

BR-KYC-001: If PEP = Yes for any director or beneficial owner → mandatory Compliance review step inserted in workflow → cannot proceed to scoring without Compliance clearance

BR-KYC-002: If AML screening = Positive → application status set to "Compliance Hold" → automated task created for Chief Compliance Officer → application blocked from proceeding

BR-KYC-003: If tax clearance certificate is expired (date > today) → validation error on submission → application cannot be submitted

BR-KYC-004: If latest management accounts are older than 180 days → warning displayed but application can proceed → note added to credit file

BR-KYC-005: If total existing debt > 3x annual turnover → automatic flag on application → Credit Risk Manager notified

3. MODULE 2 — CREDIT SCORING AND RISK ASSESSMENT ENGINE

3.1 Business Context

Fidelity Bank's current Obligor Risk Rating (ORR) model was developed in 2019 and validated by external consultants in 2022. The model is currently implemented in Excel (Version 4.2, maintained by Credit Risk Management team). The ACP configuration must faithfully replicate the ORR model logic while adding the benefits of automation, audit trails, and integration with origination data.

3.2 The Fidelity Bank ORR Model — Detailed Specification

3.2.1 Financial Ratio Component (70% weight)

The model computes the following 14 financial ratios from uploaded financial statements:

Liquidity Ratios (Weight: 15%)

1. Current Ratio = Current Assets / Current Liabilities (benchmark: >1.5)
2. Quick Ratio = (Current Assets - Inventories) / Current Liabilities (benchmark: >1.0)

3. Cash Ratio = Cash and Cash Equivalents / Current Liabilities (benchmark: >0.3)

Leverage Ratios (Weight: 20%) 4. Debt to Equity = Total Debt / Shareholders Equity (benchmark: <2.0) 5. Debt to EBITDA = Total Debt / EBITDA (benchmark: <4.0) 6. Interest Coverage = EBIT / Interest Expense (benchmark: >3.0)

Profitability Ratios (Weight: 20%) 7. Return on Assets = Net Income / Total Assets (benchmark: >5%) 8. Return on Equity = Net Income / Shareholders Equity (benchmark: >15%) 9. EBITDA Margin = EBITDA / Revenue (benchmark: >15%) 10. Net Profit Margin = Net Income / Revenue (benchmark: >5%)

Efficiency Ratios (Weight: 15%) 11. Asset Turnover = Revenue / Total Assets (benchmark: >0.8) 12. Receivables Days = (Accounts Receivable / Revenue) × 365 (benchmark: <60 days) 13. Payables Days = (Accounts Payable / COGS) × 365 (benchmark: 30-90 days) 14. Inventory Days = (Inventories / COGS) × 365 (benchmark: <90 days, N/A for service companies)

3.2.2 Qualitative Component (30% weight)

Management Quality (Weight: 15%)

- Management experience (years in industry): scored 1-5
- Management stability (turnover in last 3 years): scored 1-5
- Financial reporting quality (timeliness, auditor): scored 1-5
- Strategic clarity (business plan quality): scored 1-5

Market Position (Weight: 15%)

- Market share in primary sector: scored 1-5
- Competitive differentiation: scored 1-5
- Customer concentration risk (top 3 customers > 50% of revenue): scored 1-5
- Supplier concentration risk: scored 1-5

3.2.3 Industry Sector Adjustments

Each of the 18 sectors has a sector multiplier applied to the final score:

Sector	Multiplier	Rationale
Oil and Gas (Upstream)	0.90	High commodity price volatility
Oil and Gas (Downstream/Trading)	0.92	FX exposure, regulatory risk
Manufacturing (FMCG)	1.05	Stable demand, strong sector
Construction	0.88	Cyclical, high receivables risk
Real Estate	0.85	Illiquidity, valuation risk
Agriculture (Commercial)	1.00	CBN priority sector
Telecommunications	1.08	Strong cash flows
Financial Services	0.95	Regulatory risk

Sector	Multiplier	Rationale
Healthcare	1.02	Stable demand
Education	1.03	Recession-resistant
Retail/Trading	0.95	Thin margins, inventory risk
Logistics and Transport	0.92	Capital intensive
Technology (ICT)	1.05	Growing sector
Mining	0.88	Commodity risk
Power and Energy	0.90	Regulatory and infrastructure risk
Hotel and Tourism	0.85	High cyclicity
Professional Services	1.00	Stable cash flows
Other	0.95	Default multiplier

3.2.4 Risk Grade Mapping

Composite Score	Risk Grade	Risk Category	Recommended Action
85-100	A	Excellent	Full approval recommended
75-84	B	Good	Approval with standard covenants
65-74	C	Satisfactory	Approval with enhanced covenants
50-64	D	Marginal	Senior credit committee review required
Below 50	E	Weak	Decline recommended

3.2.5 Recommended Credit Limit Calculation

Based on the risk grade and financial analysis:

- Grade A: Up to 2.5× EBITDA or 35% of Net Worth (whichever is lower)
- Grade B: Up to 2.0× EBITDA or 30% of Net Worth
- Grade C: Up to 1.5× EBITDA or 25% of Net Worth
- Grade D: Up to 1.0× EBITDA or 20% of Net Worth (subject to Committee approval)
- Grade E: No recommendation (decline)

3.3 Functional Requirements

FR-SCO-001: The scoring engine shall automatically extract financial data from uploaded financial statements using OCR and structured data parsing. Manual override of extracted values permitted

with mandatory audit note.

FR-SCO-002: All 14 ratios shall be computed automatically and displayed with traffic light indicators (Red/Amber/Green vs benchmarks).

FR-SCO-003: The qualitative questionnaire shall be displayed to the Relationship Manager for completion. Each question presented with a 1-5 scale and descriptive anchors.

FR-SCO-004: The composite score, risk grade, and recommended credit limit shall be computed in real-time as data is entered.

FR-SCO-005: The scoring output screen shall display: composite score, risk grade, breakdown by component, ratio dashboard, and recommended limit in a single paginated report format.

FR-SCO-006: All scoring model parameters (weights, benchmarks, sector multipliers, grade thresholds) shall be modifiable by authorized Credit Risk Management users (role: Credit Risk Administrator) without IT intervention, with change approval workflow and full audit trail.

FR-SCO-007: Scoring calculations shall be reproducible — given the same inputs, the same outputs shall always be produced. Historical versions of the scoring model must be archived with date-effective records.

FR-SCO-008: The system shall support override of the computed risk grade by authorized senior credit officers, with mandatory written justification and supervisor approval for upgrades (favorable overrides) of more than one grade.

4. MODULE 3 — CREDIT APPROVAL WORKFLOW

4.1 Business Context

The Bank's Credit Authority Matrix is defined in the Board-approved Credit Policy (version 7.2, March 2026). The matrix defines lending authority by credit amount, risk grade, collateral coverage, and industry sector concentration. The current manual routing via email results in files sitting in approvers' inboxes unattended, with no visibility or escalation mechanism.

4.2 Credit Authority Matrix Configuration

4.2.1 Standard Approval Tiers

Tier 1 — Relationship Manager

- Individual authority: NGN 0 to NGN 50,000,000
- Risk grade: A and B only
- Collateral coverage: ≥100%
- SLA: 24 hours
- No committee quorum required

Tier 2 — Credit Risk Committee

- Amount: NGN 50,000,001 to NGN 500,000,000

- All risk grades (with D requiring additional justification document)
- Collateral coverage: $\geq 80\%$ for A/B, $\geq 120\%$ for C/D
- SLA: 48 hours
- Quorum: Minimum 3 of 5 committee members

Tier 3 — Board Credit Committee

- Amount: Above NGN 500,000,000
- Risk grades A, B, C only (D requires special resolution)
- Collateral coverage: $\geq 100\%$ minimum
- SLA: 72 hours
- Quorum: Minimum 4 of 7 board members

4.2.2 Special Routing Rules

BR-WF-001: If industry sector concentration $> 25\%$ of total corporate portfolio after this approval → automatic insertion of additional review step by Chief Risk Officer before board submission.

BR-WF-002: If applicant has existing non-performing facility at the Bank → workflow routed directly to Credit Risk Committee regardless of amount.

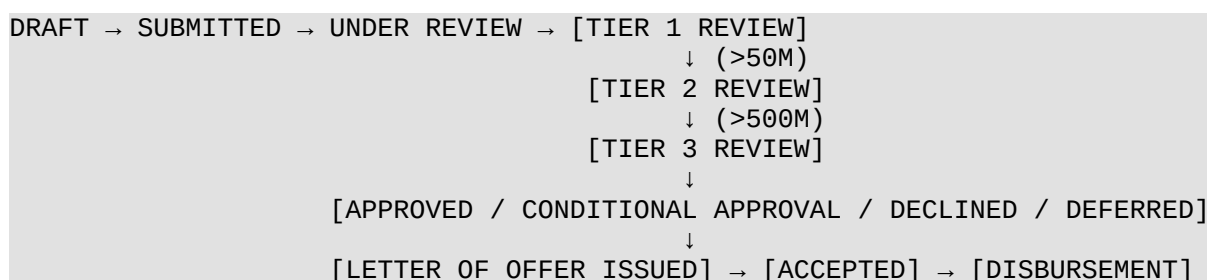
BR-WF-003: If any director or beneficial owner is a PEP → Board Credit Committee approval required regardless of amount.

BR-WF-004: Insider credit (Bank staff, related parties, directors) → automatically routed to Board Credit Committee with automatic notification to CBN (audit log only).

BR-WF-005: Restructuring of existing facilities → automatically requires one-tier higher approval than original.

4.3 Workflow States and Transitions

Each application shall pass through the following states:



4.4 Functional Requirements

FR-WF-001: The system shall automatically determine the appropriate approval tier based on the requested amount and risk grade immediately upon submission.

FR-WF-002: Each approver shall receive an in-system notification and email notification when a file enters their queue. Notification to include: client name, requested amount, risk grade, and

application reference.

FR-WF-003: Approvers shall be able to take the following actions on files in their queue:

- Approve (unconditional)
- Conditionally Approve (requires specification of conditions with due dates)
- Defer (request for additional information — mandatory field specifying information required)
- Decline (mandatory decline rationale from approved reasons list)
- Escalate (escalate to next tier, with justification)

FR-WF-004: SLA timer shall begin when file enters an approver's queue. Timer visible to approver and to Credit Operations team. Color indicator: Green (>50% time remaining), Amber (25-50% remaining), Red (<25% remaining).

FR-WF-005: Upon SLA breach, system shall automatically: send escalation notification to approver's supervisor, flag the file with "SLA Breached" status, and generate an SLA breach report entry.

FR-WF-006: The full audit trail of all workflow actions — who took what action, when, with what comment — shall be maintained and accessible in the credit file history.

FR-WF-007: Conditional approvals shall generate a checklist of conditions. Each condition can be marked "Fulfilled" by Credit Operations with supporting documentation. File shall not proceed to Letter of Offer until all conditions are marked fulfilled with evidence.

FR-WF-008: The system shall support parallel review — multiple committee members can review simultaneously, with the workflow proceeding upon quorum achievement.

5. MODULE 4 — LOAN CONDITIONS AND COVENANT MANAGEMENT

5.1 Business Context

The Bank has approximately 1,200 active covenant monitoring requirements across its corporate book. Currently managed in an Excel spreadsheet maintained by a dedicated Credit Monitoring team of 4 officers, this approach generates multiple CBN audit findings each year regarding completeness and timeliness of monitoring.

5.2 Covenant Categories and Configuration

5.2.1 Financial Covenants

The following financial covenants are standard for corporate lending:

Leverage Covenants:

- Maximum Debt to EBITDA $\leq 4.5\times$ (measured semi-annually)

- Maximum Debt to Equity $\leq 3.0\times$ (measured annually)
- Minimum Interest Coverage $\geq 2.5\times$ (measured semi-annually)

Liquidity Covenants:

- Minimum Current Ratio $\geq 1.2\times$ (measured quarterly)
- Minimum Cash Balance NGN [x] million (facility-specific, measured monthly for high-risk)

Profitability Covenants:

- Minimum EBITDA Margin $\geq [x]\%$ (facility-specific)
- Minimum Net Profit (no losses in two consecutive quarters)

5.2.2 Reporting Covenants

Standard reporting package required from all corporate borrowers:

- Monthly management accounts (within 30 days of month end) — for facilities above NGN 1 billion
- Quarterly management accounts (within 45 days of quarter end) — for all facilities
- Annual audited financial statements (within 120 days of financial year end)
- Facility utilization confirmation (monthly, for revolving credit facilities)
- Periodic insurance certificate renewals

5.2.3 Event Covenants

Borrower must notify the Bank within 5 business days of:

- Change of control (any change in ownership $>25\%$)
- Material litigation (claim $>10\%$ of net worth)
- Cross-default (default on any other obligation $>\text{NGN } 50 \text{ million}$)
- Insolvency proceedings (any jurisdiction)
- Key management change (CEO, CFO, or major shareholder)
- Dividend payment (if leverage ratio $>3.0\times$)
- Asset disposal ($>10\%$ of total assets in any 12-month period)

5.2.4 Collateral Covenants

- Annual valuation of real estate collateral by Bank-approved valuer
- Bi-annual valuation of equipment/plant and machinery
- Monthly stock monitoring for working capital facilities secured by inventory
- Comprehensive insurance maintenance with Bank noted as loss payee
- Negative pledge on major assets during facility tenor

5.3 Functional Requirements

FR-COV-001: Covenant capture screen shall be organized by category (Financial, Reporting, Event, Collateral) with ability to add custom covenants.

FR-COV-002: Each financial covenant shall have: covenant name, threshold, measurement frequency (monthly/quarterly/semi-annually/annually), measurement date, grace period (days), breach action (automated alert only / restrict drawings / accelerate).

FR-COV-003: Reporting covenants shall track: document type, due date, submission date, submitted by, document reference, status (Pending/Submitted/Waived/Breached).

FR-COV-004: Automated alert system:

- 60 days before covenant review date → notification to Relationship Manager
- 30 days before → notification to Relationship Manager + Team Leader
- 7 days before → notification to Relationship Manager + Credit Monitoring + Team Leader
- On breach → immediate notification to Credit Monitoring + Credit Risk Manager + automatic task creation for Relationship Manager

FR-COV-005: Nightly batch process shall evaluate all financial covenant measurements where measurement date falls within the next 30 days and where financial data has been updated. Breach detection logic:

- For $\leq$ type covenants: actual value $>$ threshold = breach
- For $\geq$ type covenants: actual value $<$ threshold = breach
- Grace period: if within grace period, status = "Grace Period" (not yet breached)

FR-COV-006: Covenant waiver management:

- Any waiver requires Credit Risk Manager approval minimum
- Waiver form captures: covenant name, breach description, waiver period, justification, conditions attached to waiver
- Waivers have expiry dates and generate alerts 30 days before expiry
- Waiver history maintained indefinitely in credit file

FR-COV-007: Portfolio-level covenant monitoring dashboard showing: total covenants by status (Current / Watch / Grace Period / Breached), upcoming covenant reviews (next 30 / 60 / 90 days), breach trends, sector analysis.

6. MODULE 5 — CREDIT PORTFOLIO MONITORING AND EARLY WARNING SYSTEM

6.1 Business Context

The CBN's 2020 Prudential Guidelines mandate that banks implement a forward-looking credit risk management framework, including an Early Warning System (EWS) capable of identifying

deteriorating credits before they become non-performing. The Bank's current approach relies on monthly credit review meetings and is entirely reactive.

6.2 The 12 Early Warning Indicators — Detailed Specification

EWI-01: Payment Delinquency

Trigger: Any contractual payment (principal, interest, fee) overdue by >30 days

Data Source: Core banking system (Finacle) — automated daily feed

Threshold Configuration: Configurable (default 30 days)

Severity: Red flag — immediate action required

Automated Action: Risk flag created, task assigned to Relationship Manager within 24 hours

EWI-02: Turnover Decline

Trigger: Account turnover (credits to account) declines >20% year-on-year for 2 consecutive quarters

Data Source: Account transaction data from Finacle

Threshold Configuration: Configurable (default 20%, 2 quarters)

Severity: Amber flag — enhanced monitoring

Automated Action: Enhanced monitoring tag applied, alert to Relationship Manager

EWI-03: Adverse Industry News

Trigger: Manual flag by Credit Monitoring team based on sector watch list

Data Source: Manual input with news reference

Threshold Configuration: N/A (manual trigger)

Severity: Amber flag

Automated Action: All facilities in flagged sector escalated to Credit Risk Manager review

EWI-04: Collateral Value Deterioration

Trigger: Collateral value drops >15% since last valuation, resulting in coverage falling below minimum

Data Source: Valuation updates uploaded by Credit Monitoring

Threshold Configuration: Configurable deterioration threshold (default 15%)

Severity: Red flag if coverage falls below minimum, Amber if approaching minimum

Automated Action: Valuation update request task created, top-up demand evaluation

EWI-05: Facility Utilization Anomaly

Trigger: Overdraft or revolving facility utilization consistently above 90% of approved limit for >45 days, OR utilization drops below 10% for >60 days (may indicate client sourcing funds elsewhere)

Data Source: Daily balance feeds from Finacle

Threshold Configuration: Configurable thresholds and observation periods

Severity: Amber flag

Automated Action: Relationship Manager alert, usage pattern analysis task

EWI-06: Cross-Default Trigger

Trigger: Intelligence received that borrower has defaulted on obligation to another lender

Data Source: Credit bureau (CRC, First Central) monthly feed + manual trigger

Threshold Configuration: Any default event

Severity: Red flag

Automated Action: Immediate escalation to Credit Risk Manager, review of all facilities for acceleration clause

EWI-07: Auditor Qualification

Trigger: Annual financial statements submitted with qualified or adverse audit opinion

Data Source: Document review during annual financial statement covenant

Threshold Configuration: Qualified or adverse opinion (not emphasis of matter)

Severity: Amber flag

Automated Action: Credit Risk Manager review task, request for management explanation

EWI-08: Management Change

Trigger: Notification received of change in CEO, CFO, or controlling shareholder

Data Source: Event covenant notification + manual trigger

Threshold Configuration: C-suite level only (CEO, CFO, Chairman)

Severity: Amber flag

Automated Action: Relationship Manager instructed to schedule urgent management meeting within 30 days

EWI-09: Regulatory Action

Trigger: Any regulatory sanction, investigation, or directive issued against the borrower by any regulatory authority

Data Source: Manual trigger by Compliance team

Threshold Configuration: N/A (manual)

Severity: Red flag

Automated Action: Immediate notification to Chief Risk Officer and Legal Counsel

EWI-10: Credit Bureau Deterioration

Trigger: Credit bureau score deteriorates by >15 points, or new adverse listings (judgment, repossession, administration order)

Data Source: Monthly credit bureau refresh (CRC Credit Bureau)

Threshold Configuration: Configurable deterioration threshold

Severity: Amber (score drop) or Red (adverse listing)

Automated Action: Credit bureau report task for Relationship Manager, review of facility terms

EWI-11: Covenant Breach Frequency

Trigger: 2 or more covenant breaches (financial or reporting) within any rolling 12-month period

Data Source: Covenant monitoring module (Module 4)

Threshold Configuration: Configurable frequency threshold (default: 2 in 12 months)

Severity: Red flag

Automated Action: Credit Risk Manager review, potential downgrade recommendation

EWI-12: Restructuring Request

Trigger: Formal or informal request from borrower to restructure facility terms (tenor extension, payment holiday, margin reduction)

Data Source: Manual trigger by Relationship Manager

Threshold Configuration: N/A (manual)

Severity: Red flag

Automated Action: Immediate Credit Risk Manager notification, restructuring workflow initiated

6.3 Portfolio Health Dashboard Requirements

FR-EWS-001: Real-time portfolio dashboard accessible to Credit Risk Management showing:

- Total portfolio: number of facilities, total exposure (NGN)
- Risk grade distribution: pie chart by exposure and by count
- Sector concentration: bar chart vs concentration limits
- EWI status: count of Red/Amber flags by indicator type
- Geographic distribution: map of exposures by state
- Maturity profile: bar chart of facilities maturing in 30/60/90/180/365 days

FR-EWS-002: Drill-down capability from any dashboard metric to the underlying list of facilities, with direct link to each credit file.

FR-EWS-003: The system shall generate a weekly automated portfolio monitoring report covering: new flags raised, flags resolved, sector movements, top 10 exposures, and portfolio KPIs vs targets. Report distributed by email to Chief Risk Officer, Credit Risk Manager, and EVP Corporate Banking every Monday morning at 8:00 AM WAT.

FR-EWS-004: Each EWI flag shall have a clear lifecycle: Open → Under Review → Resolved / Escalated. Resolution requires mandatory documentation of actions taken.

FR-EWS-005: Configurable concentration limits: sector, single obligor (maximum 20% of Tier 1 capital per CBN guidelines), geographic, currency. Automatic alerts when 80% of limit reached, hard block when 95% reached (requires CRO override).

7. CROSS-CUTTING REQUIREMENTS

7.1 Security and Access Control

- Role-based access control with minimum 12 defined roles (RM, RM Supervisor, Credit Analyst, Credit Operations, Credit Risk Officer, Credit Risk Manager, CRO, Committee Member, Compliance Officer, Audit, Administrator, Read-Only)
- Session timeout: 15 minutes of inactivity
- All access logged with timestamp and IP address
- Password policy: minimum 12 characters, complexity requirements, 90-day rotation

- Multi-factor authentication for all users accessing credit files above NGN 100 million
- Data encryption at rest (AES-256) and in transit (TLS 1.3 minimum)

7.2 Performance Requirements

- Credit application form: page load < 2 seconds
- Scoring computation: < 3 seconds from last input
- Workflow transition: < 1 second
- Dashboard refresh: < 5 seconds
- Report generation: < 30 seconds for standard reports
- System availability: 99.5% during business hours (8AM – 6PM WAT, Monday to Friday)
- Data backup: automated daily backup, 7-year retention per CBN requirements

7.3 Integration Requirements

System	Integration Type	Data Flow	Frequency
Finacle (Core Banking)	REST API	Account data, payment status	Daily batch + real-time for delinquency
Fircosoft (AML)	REST API	AML screening	On-demand at submission
CRC Credit Bureau	SFTP	Credit reports	Monthly refresh
First Central Credit Bureau	SFTP	Credit reports	Monthly refresh
Email Gateway (Microsoft Exchange)	SMTP	Notifications	Real-time
Active Directory	LDAP	User authentication	Real-time
Document Management (SharePoint)	REST API	Document archiving	On-demand

7.4 Audit and Compliance Requirements

- Complete audit trail for all user actions (create, read, update, delete)
- Credit decision rationale mandatory for all approvals and declines
- Scoring model version tracked with each scoring output
- 7-year data retention minimum
- Ability to reproduce exact state of any credit file at any point in time
- Audit log export capability for CBN examination
- Segregation of duties enforced: same user cannot both initiate and approve a credit application

- Four-eyes principle mandatory for facilities above NGN 1 billion
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8. IMPLEMENTATION APPROACH

8.1 Configuration vs Customization

All requirements in this document shall be met through ACP standard configuration. No custom software development is anticipated. Where configuration options are insufficient to meet a requirement, the requirement shall be formally reviewed through the Change Control process.

8.2 Data Migration

Initial data migration scope:

- Active credit files (currently in process): approximately 45 files
- Covenant monitoring schedules: approximately 1,200 records
- Existing customer profiles: approximately 3,200 records
- Historical scoring outputs (last 2 years): to be determined

Migration approach: extract from current systems, cleanse, load to ACP staging environment, validate with business users.

8.3 Testing Strategy

- Unit Configuration Testing (UCT): each configured rule tested in isolation by Axe Finance team
- System Integration Testing (SIT): end-to-end workflows tested by Axe Finance
- User Acceptance Testing (UAT): business users test real scenarios using anonymized real data
- Performance Testing: load testing to validate performance requirements
- Security Testing: penetration test by certified security firm (CBN requirement)

8.4 Training Plan

Session	Audience	Duration	Format
System Overview	All users	2 hours	Webinar
Relationship Manager Training	85 RMs	4 hours	Workshop (5 groups)
Credit Operations Training	12 staff	8 hours	Hands-on workshop
Credit Risk Management	8 staff	6 hours	Hands-on workshop
Administrator Training	3 IT staff	16 hours	Deep-dive sessions

Session	Audience	Duration	Format
Compliance Officer Training	4 staff	4 hours	Workshop

9. ACCEPTANCE CRITERIA

Each module will be accepted when the following criteria are met:

KYC Module:

- All 4 onboarding steps functional with all field validations operational
- Document upload working for all specified file types
- AML integration hook operational (mock Fircosoft response acceptable for UAT)
- 5 complete test scenarios executed without error

Scoring Module:

- All 14 ratios computed correctly vs manual Excel calculation for 10 test cases (tolerance: $\pm 0.1\%$)
- All sector multipliers applied correctly
- Risk grade and limit recommendation consistent with ORR model specification
- Override capability with audit trail functional
- Scoring model parameter modification tested with 3 changes

Workflow Module:

- All 3 approval tiers routing correctly for test amounts
- SLA timer operational with correct escalation
- All decision actions (Approve/Decline/Defer/Conditional) functional
- Notification delivery confirmed for all trigger events
- 5 complete workflow scenarios from submission to approval/decline

Covenant Module:

- All covenant types (Financial/Reporting/Event/Collateral) configurable and tracked
- Automated alert schedule operational for 30/7 day triggers
- Nightly breach detection running and accurate
- Waiver workflow functional

Portfolio Monitoring:

- All 12 EWI logic rules operational

- Dashboard displaying correct data for all metrics
- Automated weekly report generated correctly
- Concentration limit alerts functional

10. GLOSSARY

Term	Definition
ACP	Axe Credit Portal — the digital lending platform being implemented
BRM	Business Rules Management — ACP module for configuring business logic
BVN	Bank Verification Number — unique identifier for Nigerian bank customers
CAC	Corporate Affairs Commission — Nigerian company registration authority
CBN	Central Bank of Nigeria — banking regulator
DSCR	Debt Service Coverage Ratio
EBITDA	Earnings Before Interest, Tax, Depreciation and Amortization
EWI	Early Warning Indicator
EWS	Early Warning System
Finacle	The Bank's core banking system (Infosys Finacle v11.2)
KYC	Know Your Customer — regulatory requirement for client identification
NDPR	Nigeria Data Protection Regulation
ORR	Obligor Risk Rating — the Bank's internal credit scoring model
PEP	Politically Exposed Person
RM	Relationship Manager
UAT	User Acceptance Testing
WAT	West Africa Time (UTC+1)

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